

Sayan Ghorai

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Education

VIT Bhopal University

M.Tech

Artificial intelligence and data science

GPA: 7.59/10

Bhopal, Madhya Pradesh

May 2026

Haldia Institute of Technology

B. Tech

Artificial Intelligence and Machine Learning

GPA: 8.58/10

Haldia, West Bengal

May 2024

Paramanandapur Jagannath Institution

12th Standard

West Bengal Council of Higher Secondary Education

Percentage: 77.00%

Purba Medinipur, West Bengal

May 2018

Experience

Salesforce Developer Virtual Internship

November 2023 - January 2024

- Learned Salesforce Fundamentals Organizational Setup Relationship and Process Automation, Types of Flows and Security.
- Also learned Apex, Testing and Debugging VS Code Setup and CLI Setup, Lightning Web Components and API.

Projects

Multilingual Image Captioning Assistant

- Fine-tuned Florence-2 using LoRA on 50K COCO image-caption pairs for multilingual image caption generation.
- Built an end-to-end pipeline for image captioning, Bengali translation, and speech synthesis for accessibility support.
- Developed and deployed a Streamlit-based multimodal assistant using Florence-2, NLP translation, and gTTS.

GenAI Tourist Place Recommendation System

- Built a hybrid recommendation system using TF-IDF + SBERT for semantic travel search and destination ranking.
- Developed city-aware filtering, metadata-based scoring, and auto-tagging for improved recommendation relevance.
- Designed and deployed a Streamlit-based explainable travel assistant with LLM-style recommendation insights.

Plant Disease Classification

- Developed an ensemble ResNet50 + ShuffleNetV2 model for 38-class plant disease classification.
- Trained on 87K+ leaf images, achieving 99.13% validation accuracy with Grad-CAM explainability.
- Built a Streamlit web app with leaf validation, disease recommendations, and confidence scoring.

Skills

Programming Languages: C, Python

Databases: MySQL

MLOps & Deployment: MLflow, Docker, Kubernetes, Kubeflow

Libraries: Pandas, NumPy, TensorFlow, Scikit-learn, PyTorch, Transformers, PEFT, SBERT

Machine Learning: Random Forest, Naive Bayes, Regression, SVM, K-Means Clustering

Deep Learning: CNN, RNN, LSTM, Neural Network, GAN

GenAI: LoRA Fine-Tuning, Multimodal AI, Vision-Language Models, Semantic Retrieval

Tools: Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Languages: Bengali, Hindi, English

Certifications & Training:

- Python For Beginners Course In-Depth(Udemy)